M. Dhanush Kumar
192525318

CO1 - AT3 Descriptive Written Test

1) Intelligence Agents in a Smart Automation System:-

Introduction:-

An intelligent agent is a system that perceives its environment through sensors and performs actions using actuators to achieve specific goals.

1. Types of Intelligent Agents:-

a) Simple Reflex Agent

A simple reflex agent select actions based on the current percpect using predefined condition - action rules.

Ex:- when a motion sensor detects a person in a dark room, the agent switch on lights

- Advantages: Fast, simple and easy to implement.
- Limitations: Cannot Learn from past experience
- Application: Automatic Lighting and basic alerts

b) Model-Based Agent

A model-based agent maintains an internal state of the environment using current sensor inputs and past information.

Example: It remembers that a person inside a room even when the motion sensor temporarily stops

- Advantages: Handles situations where environment-observable
- Limitations: Requires memory and an accurate model.
- Applications: Occupancy tracking and temperature monitoring

c) Goal-Based Agent:

A goal-based agent selects actions that help achieve specific objective by possible outcomes.

- Advantages: Supports planning and flexible decision-making.
- Limitations: May require more computation.
- Applications: Temperature control and home security.

d) Utility-Based Agent:

A utility-based Agent evaluates different actions and selects the one that provides the greatest overall

2) Uniformed Search Strategies an Unknown Maze

Introduction:-

Uniformed search explores possible states without knowledge about the goal's location. In a craze, a robot moves through available paths until it finds the exit.

1. Uniform Cost Search (UCS)

Uniform cost search expands the node with the lowest cumulative path cost (g(n)).
It uses a priority queue to select the least-cost path.

Advantages:

- Finds the minimum-cost path when steps costs are non-negative and the search is complete.
- Handles different movement costs.
- Avoids selecting an expensive route when a cheaper route is available.

Complexity:- For a finite branching factor b, solution d and equal positive step costs, the worst-case time and space complexities are Complexity O

2. Iterative Deepening Search :

Iterative Deepening search repeatedly performs depth-limited depth-first Search.

Working procedure:-

1. Begin with depth zero.
2. Search the maze within the limit.
3. if the exit is not found, increase limit by one.
4. Stop when the exit is reached.

Advantages:-
- uses less memory than UCS.
- is complete when the branching factor is finite

Complexity:
- Time : $O(b^d)$
- Space : $O(bd)$

3. Comparisive of UCS and IDS

Feature	UCS	IDS
• search method	lowest cost first	Increasing depth limit
• Data structure	Priority queue	DFS stack
• completeness	under suitable.	Yes, finite branching.

Optimality	minimum - cost Path	Shallowest Path for Equal Costs.
Memory	High	low

4) Role of Intelligent Agents in Maze Navigation

- Simple Reflex agent: chooses movements based on current sensor information.
- Model - Based Agent: Remembers visited cells and discovered walls to avoid repeated exploration.
- Goal - based agent: selects actions that help the robot reach the exit.

Conclusion :-

UCS is effective when minimizing path costs is important, while IDS is suitable when memory is limited and movement costs are equal.